Description of the laboratory work No. 4

TIMERS/COUNTERS AND ADC CONVERTERS OF 32-BIT MICROCONTROLLERS

1. Objectives
2. Preparation for the work
3. Workflow
   3.1. Work with a teacher
      3.1.1. STM32CubeIDE
      3.1.2. Template of the code
      3.1.3. Main functions
   3.2. Work by individual task
   3.3. Literature

1. Objectives

Get to know the architecture, functionality and configuration features of the timers/counters and ADC converters of the 32 bit microcontrollers. To develop abilities to analyze microcontrollers specifications. Get to know the software, which is needed for program of 32 bit microcontrollers.

2. Preparation for the work

We will work with timers/counters and ADC converters of 32-bit microcontrollers during this laboratory work. Students will try to accomplish several tasks independently after getting to know the basics.

3. Workflow

3.1. Work with a teacher

The lecturer demonstrates boards, explains their components and students start to program individually.

3.1.1. STM32CubeIDE

The detailed configuration of the project has already been discussed in the description of the first three laboratory works, so only the essential print screens of the graphical configuration windows that are required for the laboratory work no 4 are presented here. Print screens are presented without detailing them. These are different options you will need in the different variants of configuration. The purpose of these print screens is to prompt students about the information where the necessary configurations can be found.



Fig. 1. PWM Generation CHx“ mode. Pinouts

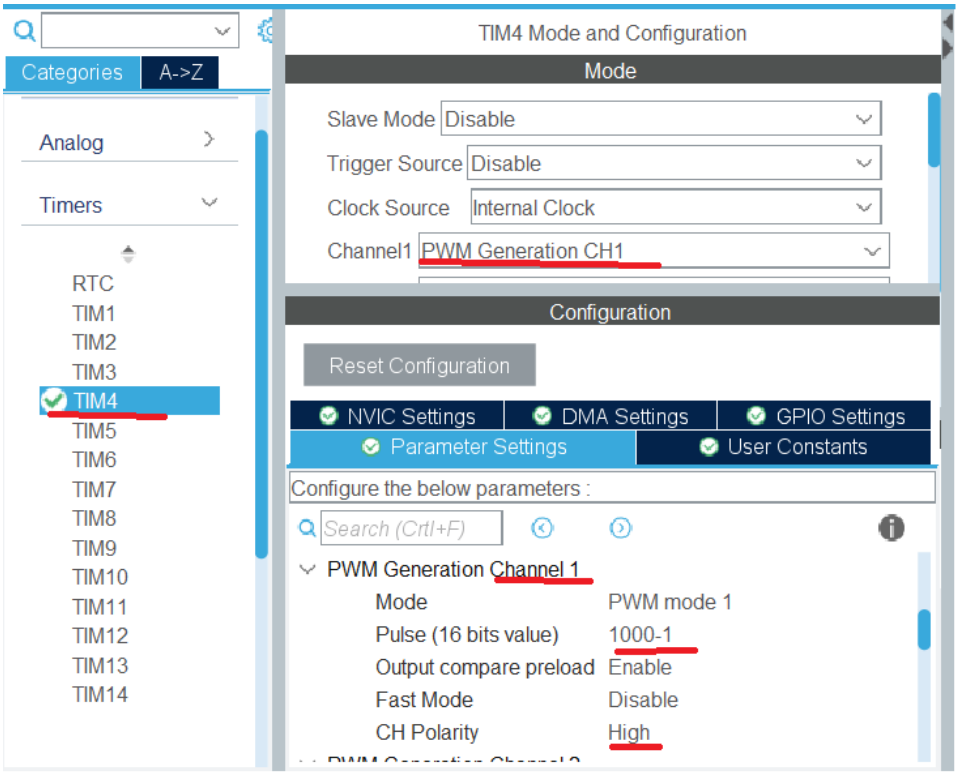


Fig. 2. „PWM Generation CHx“ mode. Configuration parameters



Fig. 3. „Input Capture Direct Mode“ mode. Pinouts

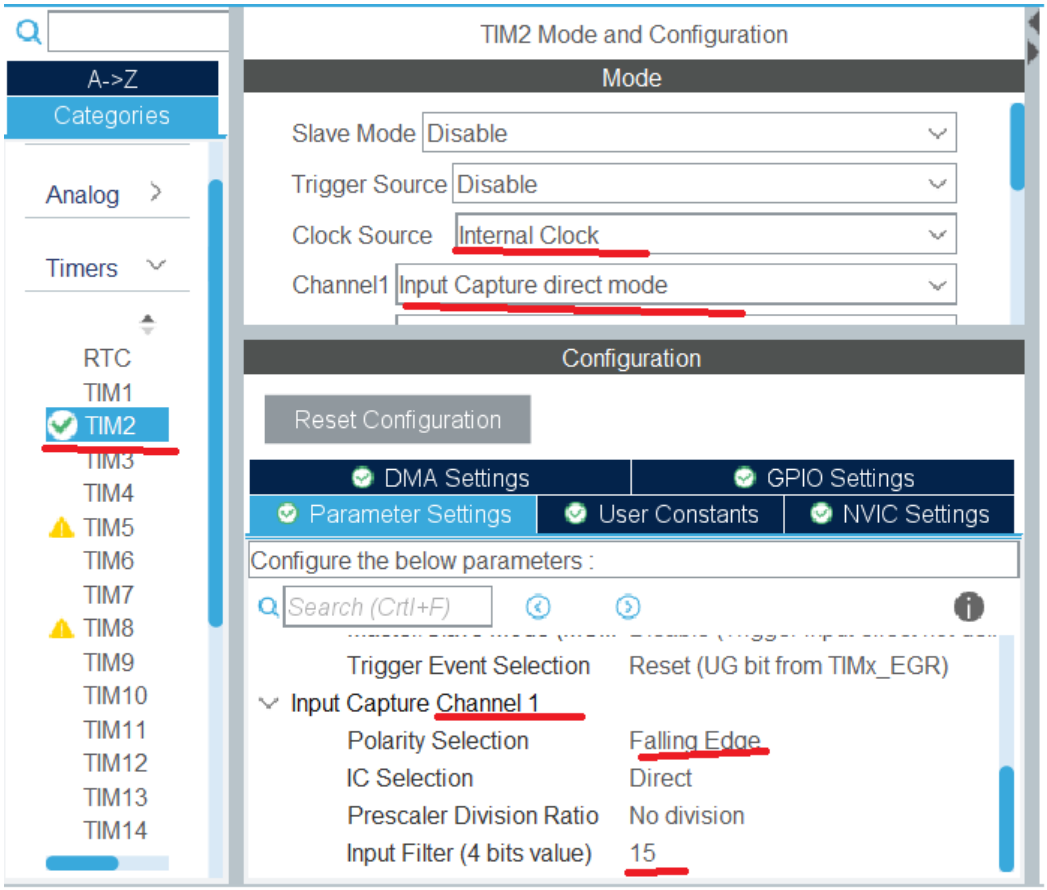


Fig. 4. „Input Capture Direct Mode“ mode. Configuration parameters



Fig. 5. „ADC“ mode. Pinouts

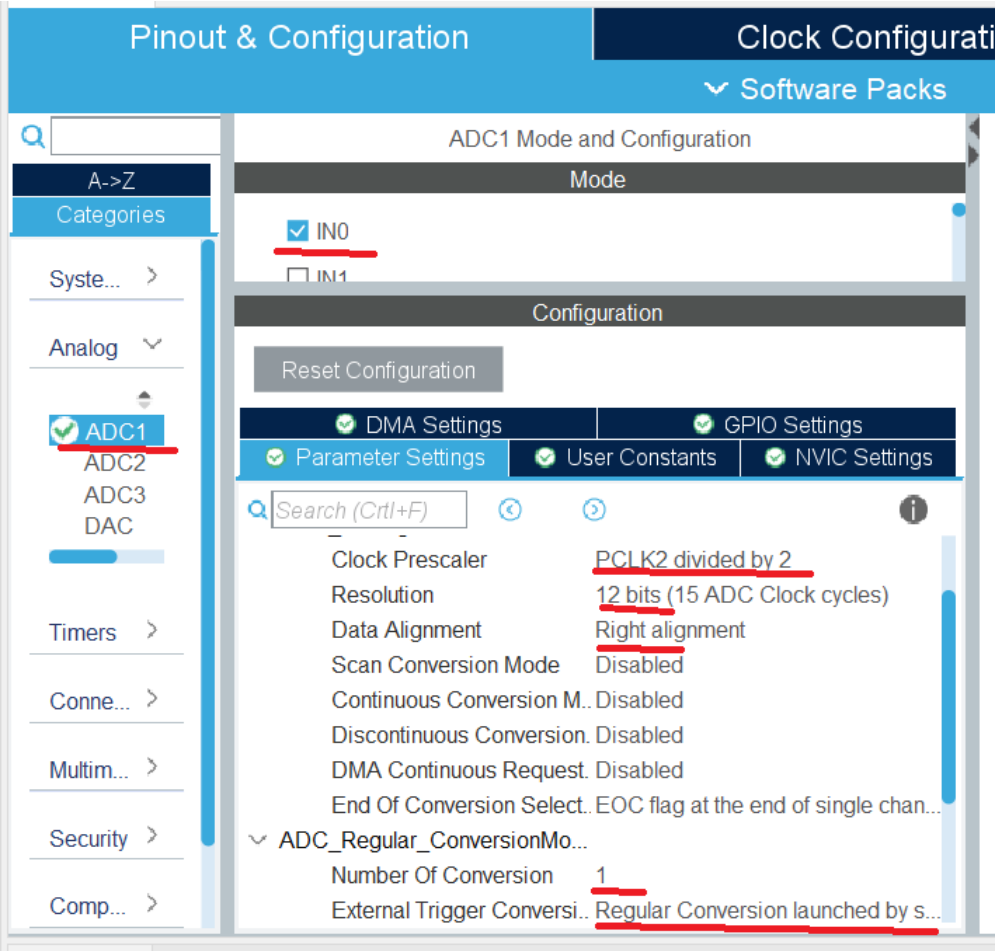


Fig. 6. „ADC“ mode. Configuration parameters

Note. These are generalized examples of configuration. It will be necessary to adjust the parameters in every specific variant of the task.

3.1.2. Template of the code

Examples of the code templates are provided in the descriptions of previous laboratory works. I just remind that it will be necessary to work with the *main.c* and *stm32f4xx_it.c* files, depending on your variant of the task.

3.1.3. Main functions and registers

- HAL_TIM_IC_Start_IT(…) – Non-Blocking mode: TIMx Interrupt.
- __HAL_TIM_GET_COUNTER(…) – Get the TIM Counter Register value on runtime.
- __HAL_TIM_SetCounter(…) - Set the TIM Counter Register value on runtime.
- TIMx_IRQHandler(…) - Function handles TIMx global interrupt.
- HAL_TIM_IC_CaptureCallback(…) - Input Capture callback in non-blocking mode.
- HAL_ADC_Start(…) - Enables ADC and starts conversion of the regular channels in blocking mode.
- HAL_ADC_PollForConversion(…) - Poll for regular conversion complete.
- HAL_ADC_Start_IT(…) – Enables the interrupt and starts ADC conversion of regular channels.
- HAL_ADC_ConvCpltCallback(…) - Regular conversion complete callback in non blocking mode.
- ADC_IRQHandler(…) - This function handles ADC1, ADC2 and ADC3 global interrupts.

3.2. Work by individual task

Students independently perform individual work assignments provided by the lecturer. Students can use any additional literature, but cannot talk to each other. When faced with difficulties and finding no solution, students can always ask specific questions to the teacher.

3.3. Literature

1. Slides of the lectures in the moodle.vgtu.lt system.
2. Code examples in the internet.
3. STM32F407. In the moodle system.
4. STM32F407 Discovery. In the moodle system.
5. UM1467 Getting started with software and firmware environments for the STM32F4DISCOVERY Kit. In the moodle system.
6. UM1472 Discovery kit for STM32F4. In the moodle system.
7. DM00031020 stm32f4x reference manual registers level. In the moodle system.